SOME RESULTS OF THE DEVELOPMENT OF A FAST REACTOR FUEL ELEMENT*

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1. Introduction

Within our fast breeder project a major effort was started some years ago to develop fuel elements suitable for large fast power reactors with different types of coolant. While primarily three coolant mediums, namely sodium, helium, and steam, were considered the project has been concentrated now to the two lines of sodium or steam cooling, respectively. In the first step of development also different possible fuel types, that is metal, oxide, and carbide were evaluated with the conclusion that only ceramic fuel may meet the economic requirements which are highly emphasized in the project. Within the different ceramic fuels mechanically mixed oxide was finally chosen as the fuel for the prototype reactors. This decision is mainly based on the available technical background and experience in the ceramic field. Although our present initiative is concentrated only on oxide fuel we may try a second approach in the future with other ceramics.

The present paper describes shortly the logical structure of our work, the experimental activities including some available results and the theoretical background we try to establish for the whole fuel element development.

2. Objectives and Programme

Requirements

The requirements for our fuel element development are based upon the fast breeder reactor design studies for a 1000 MW(e) sodium cooled reactor and for a 1000 MW(e) steam cooled version. Although the basic parameters of both designs have a good deal of similarity there are two different development lines. Some of the layout parameters are exactly equal for both types while other requirements—for instance, corrosion performance or mechanical

* This paper has been prepared within the framework of the association Euratom-Gesellschaft für Kernforschung mbH. in the field of fast breeder development.

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features of the cladding—are quite different. In any case a high burnup (up to 100,000 MWd/t) is required. The maximum linear rod power in the present reactor designs lies between about 400 W/cm (steam cooling) and 600 W/cm (sodium cooling).

Fundamental Working Hypotheses

In order to profile a reasonable development programme it is necessary to establish some working hypotheses, especially for the fuel parameters. These assumptions are step by step streamlined following the evaluation of our own development work and the results in the published literature. They deal with the following groups of fuel pin parameters:

(a) fuel density, expansion and swelling;
(b) fuel temperature distribution;
(c) fuel composition and fuel-clad interaction;
(d) pressure build-up and cladding design.

The detailed working assumptions are as follows:

1a. The basic requirements in view of solid fission product swelling and slumping danger are met in the density region 80–90% theoretical density (smeared), presumably at about 85% th.d.

2a. The thermal expansion of porous fuel ceramic is similar to 100% dense material; therefore we take the (conservative) 100% values for calculating the necessary gap width.

3a. The model for radial swelling assumes an initially low swelling rate increasing continuously with burnup.

4a. With He-bonding and a diametral gap of less than 150 μ the thermal gap conductance is not worse than 1 W/cm^2 °C. A corresponding temperature rise is also applied to vibratory compacted fuel, thus counterbalancing the lower conductivity of powder fuel in the cooler regions.

5b. The thermal conductivity integral (from 0°C to melting point) for the 100% dense mixed oxide is not lower than 95 W/cm. For the reduction due to porosity a conservatively modified Loeb-equation is applied.

6b. At local fuel temperatures above 1600°C migration of voids is assumed. The central hole volume thus formed will be not larger than about 50% of the original porous volume.

7c. Concerning the stoichiometry question we assume that stoichiometric or only slightly substoichiometric oxide can reliably be fabricated and do not change the composition during burnup in an amount that a larger adverse influence to related parameters (as thermal conductivity) must be expected.

8c. The necessary microscopic homogeneity in the UO_2–PuO_2 mixture can properly be reached by mechanical mixing. Provided that the maximum excursions have ramp rates below \$15/s—as evaluated in our Na1 reference design study[1]—theoretical calculations show that the largest highly Pu-concentrated oxide spots should be smaller than 150 μ in diameter.[2] A slight macroscopic segregation trend—for instance, induced by diffusion in a thermal gradient—will not be harmful to fuel behaviour or core physics. Also some few PuO_2 agglomerates above 150 μ are insignificant.

9c. The compatibility between oxide fuel and the proposed can material categories (stainless steel and Ni-base alloys) will raise no major problems if only He-bonding (or vacuum) is applied.

10d. For calculation of fission gas pressure the fission gas release is (conservatively) assumed to be 100% thus counterbalancing some possible drawback coming from the doubt whether the gas can always migrate completely free to the plenum.

11d. The most important characteristic property of the can material is the long term creep behaviour which has to face the superposition of gas pressure stress and thermal stress. The yield point may be surpassed only in the first few thermal cycles.

12d. Properly evaluated single alloys of the chosen can material categories suffer no major detrimental effects of high fast neutron integrated flux to their overall mechanical behaviour.

A lot of these hypotheses are highly conservative or to some extent founded by experimental results while others are only theorized and need practical proof.

Structure of Development Programme

The whole development programme began with the evaluation of the specific requirements and led in a first step to some fundamental decisions concerning pin type and fuel, see the formal scheme in Fig. 1. On this working basis a network of development work concerning the fuel, can material, and fuel pin production was established followed by an irradiation programme concentrated not only on fuel and single cladding materials irradiation but also on performance tests of prototype fuel pins. In the last step all the experimental results will be used to formulate the concept, design, and specification of a prototype fuel element for both lines of development.

3. Experimental Development Work

Test Production of Fuels

In preliminary steps the proposal to increase the thermal conductivity of the oxide ceramic by some metal additives was experimentally evaluated. Some encouraging results were reached by Mo-addings of 10–15 wt.%, uniformly distributed or as particle coatings. The normally low thermal conductivity at temperatures above 1000°C jumped to doubled or threefold values. In view of this not too large progress compared to the additional production difficulties, but especially in view of the detrimental influence of any Mo in the core to the fast reactor physics (Doppler and coolant void coefficient, breeding ratio) we have dropped this fuel side line completely.

All our effort is focused now on pure oxide where we distinguish three simulation steps, namely UO_2, UO_2–CeO_2, and UO_2–PuO_2, both mixtures mechanically mixed. Owing to some lack in Pu facilities in the first phase of

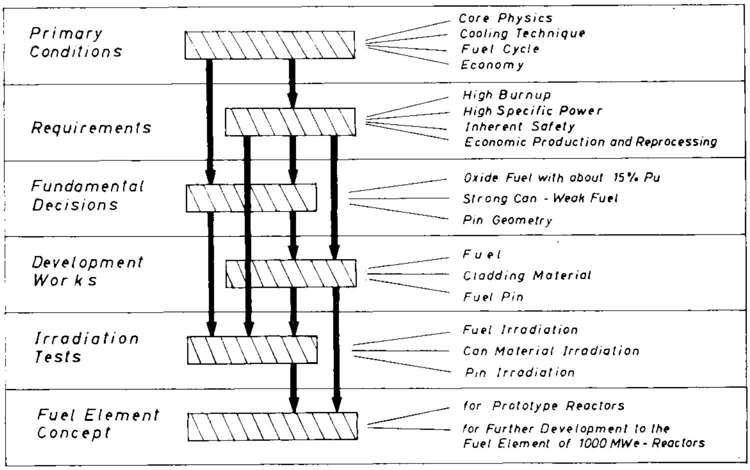


FIG. 1. Structure of fuel element development.

development we had to gain the preliminary know-how with simulated fuel. During the last year, however, we changed step by step to the handling of Pu-containing oxide.

In all our work the two basic fuel types, that is pellet fuel and compacted powder, are examined in parallel lines. For the pellet type we realized the production of small diameter pellets (5–7 mm) with densities in the range 89–92% th.d. We produced rather large series in the order of 10,000 pieces, mostly without dishing. Thus we got some production routine experience with such small pellets. The homogeneity as reached by mechanical mixing is sufficient. Large series of pellets can be sintered with such precision that they need no final grinding procedure if a diameter tolerance of $\pm 50\ \mu$ is specified.

For the vibration powder production a new method was evaluated avoiding the large and costly feedback of sintered oxide refuse. In this new process the ceramic grade material is pressed to large green pellets and milled before sintering to particles. After that the green particles are rounded, screened to a proper vibration characteristic, and sintered. We attained with powder at 5·5–7 mm fuel diameter a density up to 87% of theoretical routinewise. We are also earnestly considering the powder production process by melting in direct current, which was developed by Nukem recently.[3] This is a very easy way to produce high density particles on a large scale. The vibration results are very encouraging, too. It seems that no major difficulties for such a routine production with Pu may arise.

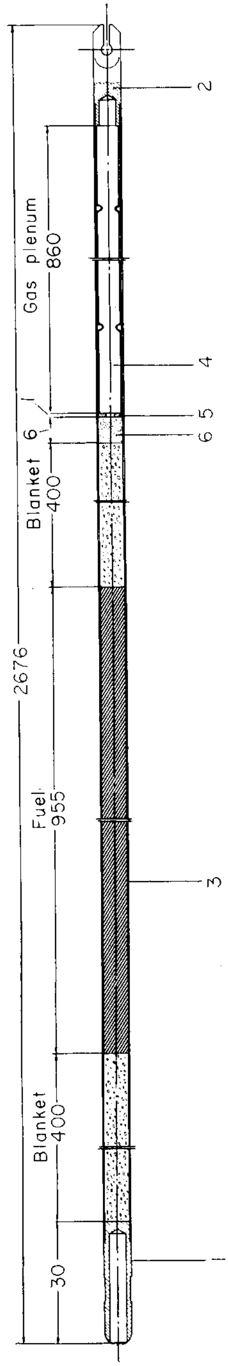
These different oxide lines are rather promising especially from an economic standpoint. Our present effort aims to verify this conclusion—based on experience with UO_2 and UO_2–CeO_2—also for Pu-containing mixed oxide Therefore we are about to enlarge the laboratory scale with UO_2–PuO_2 to a real test production scale.

Cladding Materials

The strong can concept which is exclusively pursued in this stage of development asks for cladding materials with good creep behaviour at high temperatures. A characteristic maximum temperature (including hot spot conditions) for Na cooling is 650°C, for steam cooling 750°C. At these temperatures a long-term mechanical strength of at least 15 kg/mm^2 should be available. This goal can be envisaged—from a principle standpoint—with different classes of alloys. Of course, the commercially advanced Fe-base and Ni-base alloys have to be considered first, while other promising classes, in particular vanadium and its alloys, need a long range development.

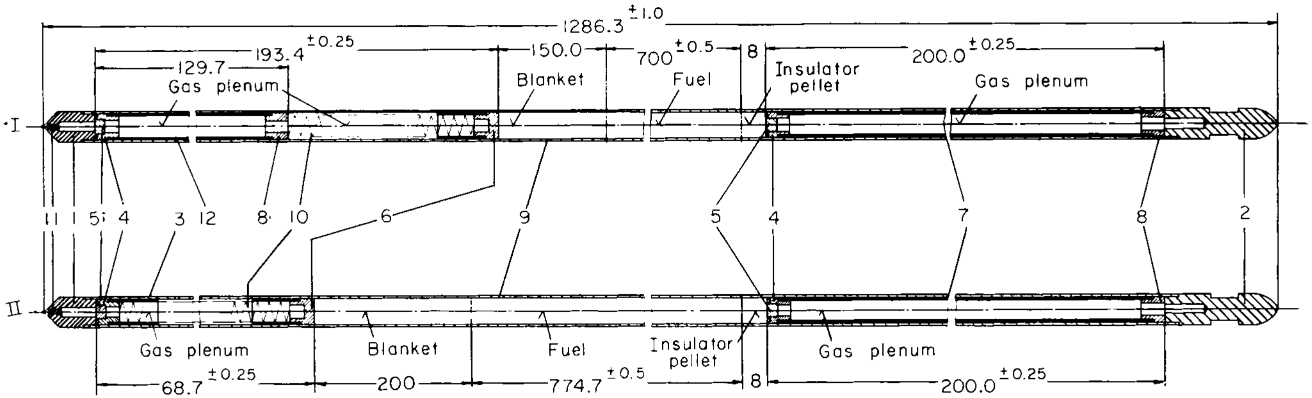
We have investigated in screening tests a lot of the different stainless steels and the available Ni alloys of the Inconel type. Our present selection is mainly concentrated to a 16Cr13Ni stainless steel for Na cooling and Inconel 625 for steam cooling. For V-base alloys some years ago we started a metallurgical development. The already completed first stage of experimental work investigated the possible alloying components and the mechanical properties of such compositions. Now the production studies for tubes of small diameters are under way, it is possible to fabricate tubing of V–10Ti and V–10Ti–10Nb. The development of more advanced Fe-base alloys, especially of the Incoloy type, is considered to be very promising. While the presently available Incoloy 800 is still not high enough in mechanical strength we expect a marked improvement with some additives. We are stimulating such development lines and hope to get further can material for steam and sodium cooled projects.

On all these potential cladding materials the most important property data are rechecked by our own out-of-pile testing. The mechanical experiments deal mainly with short and long time creep rupture tests (up to 25,000 hours) at high temperatures. Also some corrosion experiments in sodium and steam were carried out. The preliminary conclusion is that for sodium cooling 16Cr13Ni stainless steel and for steam cooling Inconel 625 meet the mechanical and corrosion requirements best.



6	Filter pellet
5	Plug
4	Insert tube
3	Cladding
2	Lower end cap
1	Upper end cap

FIG. 2. Pin design for test production.



Teil	Stück	Benennung		Werkstoff Typ I	Werkstoff Typ II		Bemerkung
12	1	Gasrohr	insert tube	16/13 Cr Ni Nb			
11	2	Stift	pin	16/13 Cr Ni Nb	Inconel 625		
10	2	Feder	spring	S.S.	Ni-base alloy		
9	2	Hüllrohr	cladding	16/13 Cr Ni Nb	Inconel 625		
8	3	Stopfen	plug	16 13 Cr Ni Nb	Inconel 625		
7	2	Gasrohr	insert tube	16 13 Cr Ni Nb	Inconel 625		
6	2	Stopfen	plug	16 13 Cr Ni Nb	Inconel 625		
5	4	Sinterstahlfilter	SS filter				
4	4	Stopfen	plug	16 19 Cr Ni Nb	Inconel 625		
3	3	Führungsrohr	sliding tube		Inconel 625		
2	2	Verschluß	lower end cap	16 13 Cr Ni Nb	Inconel 625		
1	2	Spitze	upper end cap	16 13 Cr Ni Nb	Inconel 625		
Maßstab 2:1		Irradiation test-pin for Enrico Fermi-Reactor				TA 2K-3-07-02-1395	

FIG. 3. Pin for EFFBR irradiation (types I and II).

Test Production of Fuel Pins

This is the final step of the experimental development work. It uses fuel produced according to the production routine and cladding materials of the selection described. There are two different objectives. On the one side we have to establish and to try a realistic pin production on a larger scale in order to supplement pure laboratory knowledge by some more technical experience. On the other side irradiation specimens are to be produced for the fuel irradiations programme of section 4.

The large scale pin production uses as simulation fuel natural UO_2. The pin design is according to one of the reactor reference designs (see Fig. 2). The production comprises about 200 pins. Each set of about twenty-five pins is slightly different from the others, to get experience in the different fuel types. The results of this test production show whether the established specifications can be met and give also some valuable indication concerning the production costs.

The fabrication of irradiation specimens is a more delicate task, because the specifications are more rigid than for routine production. About 100 specimens have been fabricated up to now. Presently the pins for the irradiation in the EFFBR are in preparation. As Fig. 3 demonstrates they show all important features (including the dimensions) of real oxide breeder fuel pins.

Economic Consideration

Our test productions of fuels and pins were used to establish economic outlines for a future real large scale production. The requirement was expressly not to figure out the present situation but the extrapolated state, say, at a continuous production of about 20 tons of fuel per year after having overcome initial difficulties in a start-up period. We are presently calculating the pin production costs (per kilogram of fuel) in oxide fuel type. We shall deal with the results of this investigation in a later publication.

4. Irradiation Experiments

Experimental Questions

The fundamental question is directed to the relation between obtainable burnup of a fuel pin on the one side and production specifications and operation conditions on the other side. In this connexion the fuel density and the linear rod power are most important. The definition of the expression obtainable burnup has at least two increments. We define a special burnup to be obtainable if:

(a) on the outside of the pin no major change, geometrical or otherwise, has come about which prohibits or limits normal function;
(b) on the inside of the pin no conditions are being established or developed which influence adversely normal function or which have to be considered as a potential danger for reactor safety.

Within this framework, all special questions which can be treated by irradiation experiments are included, e.g. change of temperature distribution, central hole formation, radial and axial swelling, migration of fuel components and fission products. Above that a special emphasis is justified for the question whether detrimental changes and influences are developed continuously or whether there exist critical periods in the life of a fuel pin. One such critical period is the start-up of newly produced fuel. Therefore the start-up behaviour of a pin requires additional attention. Another such critical period is an intermediate swelling stage at which the fission gas remains trapped in the fuel.

Scope of the Irradiation Programme

The whole irradiation programme is divided into different test series. Some deal only with the irradiation of can material specimens in a hard neutron spectrum and constitute the material irradiation programme. Parallel to this the fuel irradiation programme which contains short fuel specimens and complete fuel pins is carried out, most of it in a thermal neutron spectrum, a very important part however in a fast flux. Tables 1 and 2 show a compilation of the whole programme and also give the most characteristic features of each test series.

Naturally the irradiation space with a real fast flux is very limited. Therefore it is highly important that many experiments also give valuable results in thermal reactors for a fast reactor development. In this situation the question of how to profile the experiments and to distribute them on to the available irradiation space arises. The following gives the answers for our programme:

(a) as the mechanical properties of can materials are highly influenced by fast neutrons the materials irradiation take place in a high flux region of a hard spectrum. Thus integrated fast fluxes up to 5×10^{22} nvt are obtainable within reasonable time;
(b) most of the fuel and fuel pin irradiations can be undertaken in thermal reactors as there should not be large differences between thermal and fast flux in the fuel behaviour;
(c) finally one series of pin irradiations is carried out in a real fast reactor as a performance test and as a real proof for the assumptions.

TABLE 1. FUEL IRRADIATION PROGRAMME

	Test group	Number of specimens	Fuel diameter (mm)	Specimen length (mm)	Fuel type, pellets + vibro-compacted powder	Cladding temperature (°C)	Linear rod power (W/cm)	Maximum burnup (MWd/t)
Capsule irradiations in FR 2	1	20	10·0	240	UO_2, UO_2–Mo, UO_2–CeO_2	500	450	7000
	2	32	8·6	240	UO_2, UO_2–CeO_2	500	500	9000
	3	64	6·4	173	UO_2, UO_2–CeO_2	520–580	700	80,000
	4	32	6·4	173	UO_2–PuO_2	520–580	700	100,000
	5	8	5·55	1300	UO_2–PuO_2	520–580	650	100,000
Loop irradiations in FR 2	1	24	8·6	114	UO_2	550	700	5000
	2	26	10·0	160	UO_2, UO_2–CeO_2	600	1000	—
	3	26	10·0	160	UO_2–PuO_2	600	1000	—
	4	20	8·6	114	UO_2–PuO_2	550	700	50–100,000
Pin irradiations in EFFBR		36	5·55	1286	UO_2–PuO_2	550	600	60,000

TABLE 2. MATERIAL IRRADIATION PROGRAMME

Specimen type and reactor	Test designation	Materials	Number of specimens	Irradiation conditions			
				Contact medium	Temperature (°C)	Maximum integrated flux	
						Thermal (nvt)	>0·1 MeV (nvt)
Tensile specimens in BR 2 and FRJ 2	Mol 1	16Cr13Ni; 15Cr25Ni; 20Cr25Ni; Inconel 102, 600, 625, 718, X750; Incoloy 800; V-alloys; TZM	473	H_2O	< 100	$1{\cdot}5\times10^{21}$	$1{\cdot}4\times10^{21}$
	Mol 3	16Cr13Ni; Inconel 625, 718; V-alloys	72/50	He/Na	600	5×10^{21}	5×10^{21}
	DIDO M	16Cr13Ni; Incoloy; Inconel 625, 718; V-alloys	96	He	600	1×10^{21}	5×10^{20}
Pressurized tubes in BR 2	Mol 2	16Cr13Ni; 20Cr25Ni; Incoloy 800; Inconel 625, 718, X750; V-alloys	112	He	600/700	$2{\cdot}5\times10^{21}$	$2{\cdot}2\times10^{21}$
Tensile specimens in EFFBR	Fermi M	16Cr13Ni; Inconel 625; V-alloys	64	Na	350		5×10^{22}

Fuel Irradiations

The fuel irradiations of our programme can be divided into three large sections:

(A) The capsule irradiations in the FR 2 reactor.
(B) The irradiations in the FR 2 helium-loop.
(C) The irradiations in an EFFBR high flux position.

(A) For irradiations on normal fuel element positions of the FR 2 reactor in Karlsruhe a special capsule was developed. In the latest design the heat transfer between fuel and cooling water (D_2O) is performed by a twin layer of liquid metals, namely Na and Pb–Bi alloy. In such a capsule four short specimens with a length of about 17 cm each can be installed or one long specimen which has good similarity to realistic dimensions. The surface temperature of the specimens is controlled by thermocouples. The construction allows a linear rod power of up to 750 W/cm. With this device five main test groups are specified. Test group 1 contained five capsules and twenty short specimens. The initial irradiation of this type was carried out at a rod power of 450 W/cm maximum up to about 7000 MWd/t burnup. Test group 2 with eight capsules is irradiating thirty-two specimens with reduced diameter at somewhat higher rod power. Test group 3—now under way—will still contain a rather large number of short specimens. It constitutes a screening test which compares different fuel variants at different conditions up to very high burnups. Test group 4 which repeats a similar screening with Pu-containing fuel is a very important supplement to test group 3. Test group 5 now will bring the first long specimen irradiations in the capsules. The pins will be designed similar to the EFFBR pins, see below. Further test groups with capsules are planned but not yet specified in detail. They are reserved for additional fuel variants and new concepts.

(B) For the central channel of the FR 2 a He-loop with two irradiation devices was designed and constructed. The long term irradiation rig operates at rod powers up to 700 W/cm at 8·6 mm fuel diameter. The irradiations now under way lead up to 100,000 MWd/t burnup under these gas cooling conditions. With the short term irradiation rig the test samples can be introduced into the reactor core and removed during reactor operation. Thus short term changes of the fuel, especially the start-up behaviour, can be observed. The short term irradiation programme comprises about sixty specimens which will be irradiated for short periods between minutes and days at rod powers between 500 and 1000 W/cm.

(C) Under an APDA–Euratom contract one high flux position will be available in the Fermi reactor in the U.S.A. for our programme. The preparations envisage the irradiation of two or three batches of twelve pins each. For the pin design see Fig. 3. Each batch will contain pins with different cladding materials (16Cr13Ni SS and Inconel 625) and different oxide fuels (pellets and vibro-compacted). The rod power will be about 600 W/cm at a maximum sodium temperature of 550°C. The pin design is based on the assumptions for 100,000 MWd/t burnup but because of contractually limited irradiation time we shall not reach more than about 60,000 MWd/t.

Examination of Irradiated Fuel

For the post-irradiation examination of fuel specimens and pins we follow a working scheme which contains all important steps (see Fig. 4). Especially the microprobe technique is included.[4] Each investigation is accompanied by a thorough calculation concerning the obtained accuracy in the light of the experimental possibilities.

Material Irradiations

As most of the fuel irradiations are not fully realistic concerning cladding behaviour, the material irradiation programme must try to simulate the requirements step by step. It has three major sections:

(A) irradiation of tensile specimens in a thermal reactor with high fast flux component at low and high temperatures (Mol 1, Mol 3, DIDO M);

(B) irradiation of pressurized tubes in a thermal reactor with high fast flux component at high temperatures (Mol. 2);

(C) irradiation of tensile specimens in a fast reactor (Fermi M).

(A) The cylindrical and flat specimens of the test series Mol 1 are irradiated in the reactor BR 2 at Mol, Belgium, at temperatures below 100°C in direct contact with the cooling water. In this very simple parameter constellation it is still possible to study some of the most important features of the irradiation behaviour of structural materials. First the influence of fast neutrons on low temperature embrittlement becomes evident. Secondly, the high temperature embrittlement induced by (n,α) reactions can be evaluated. Finally, the annealing of low temperature damage, especially the radiation annealing temperature, can be determined in detail applying different annealing schedules after irradiation. We have carried out many tests of this type (Table 3). As Fig. 5 for some of the investigated materials demonstrates the low temperature embrittlement after an integrated fast flux of about 3×10^{20} nvt (>0·1 MeV) can be fully annealed. The results of another test, however, show that this is not completely possible at a higher irradiation dose. Examples for high temperature embrittlement are demonstrated in Fig. 6. The details of these investigations are published elsewhere.[5] Also some related studies

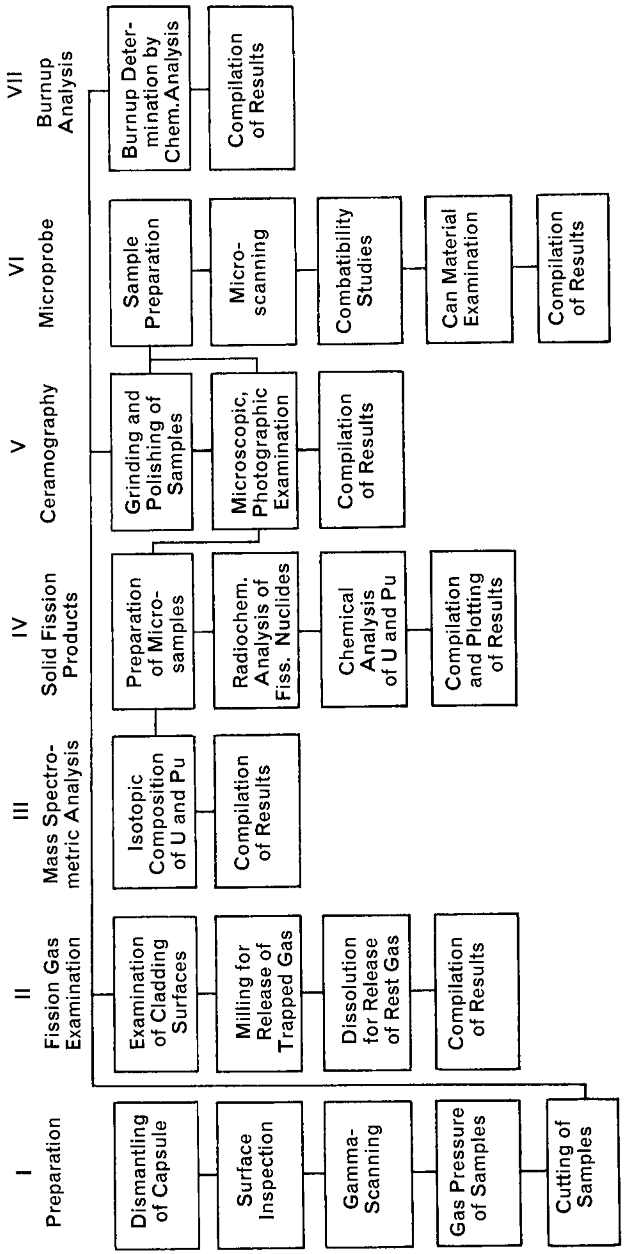


FIG. 4. Scheme for post-irradiation examination.

background[7] and with the irradiation behaviour of precipitation hardened alloys[8] were carried out. The Mol 1 series is being continued, recent additional results concerning the irradiation of V-base alloys are presently compiled.[9] The post-irradiation examination of further tests will also include long term creep experiments after the proper creep devices are installed in

TABLE 3. SPECIMENS IRRADIATED IN THE Mol 1 SERIES

Test	Material	Number of specimens	Average integrated flux in 10^{20} nvt	
			Thermal	>0·1 MeV
Mol 1A	16Cr13Ni 15Cr25Ni Inconel X750	16 15 30	5·5	4·8
Mol 1B	16Cr13Ni 20Cr25Ni Inconel 600 Inconel X750	15 15 15 16	3·0	2·7
Mol 1C	16Cr13Ni 20Cr25Ni Inconel 600 Inconel X750	15 15 15 16	13·0	11·0
Mol 1D	Inconel 102 Inconel 625 Inconel X750 Incoloy 800 V-base alloys	8 3 24 1 45	16·5	15·0
Mol 1E	V-base alloys TZM	30 11	17·0	15
Mol 1F	Incoloy 800 Inconel X750	31 50	17·6	16
Mol 1G	16Cr13Ni Inconel 625 Inconel 718	30 29 28	In preparation	

our hot cell facilities. The Mol 3—and the DIDO M—test series (in the reactor FRJ 2, Jülich) work with the same type of specimens as Mol 1 but at high temperatures of 650°C in different contact media as inert gas and sodium. With the right temperature (and possibly the real cooling medium) at the specimen's surface these experiments are some steps closer to reality. In addition to the results of the Mol 1 tests, an exact and final conclusion on high temperature embrittlement needs an experimental evaluation of the dealing specifically with high temperature embrittlement[6] and its theoretical influence of the irradiation temperature on embrittlement. After irradiation the Mol 3 specimens will mainly be examined in creep tests up to 3000 hours in accordance with the long term stress in a fuel pin.

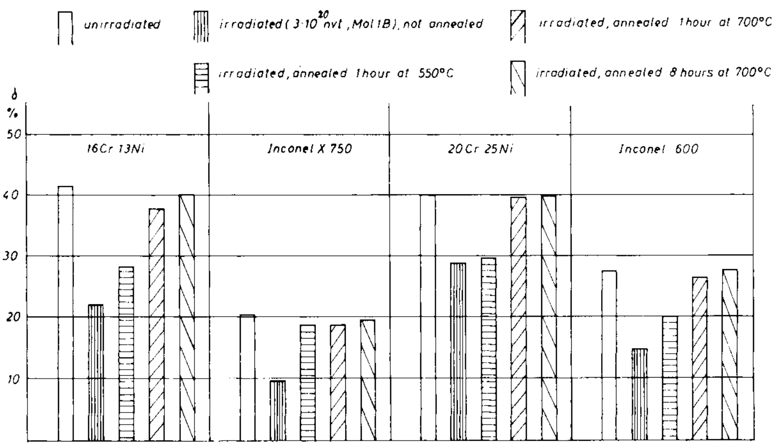


FIG. 5. Low temperature embrittlement: total elongation at room temperature.

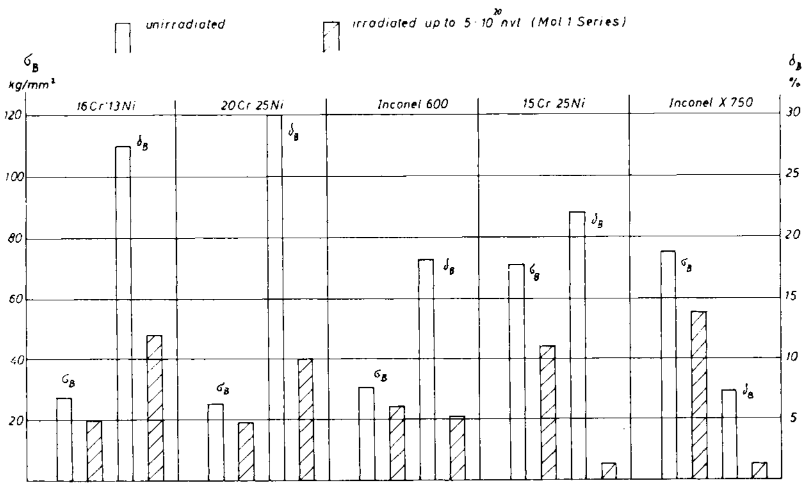


FIG. 6. High temperature embrittlement: tensile strength and total elongation at 750°C.

(B) The pressurized tubes bring remarkable additional steps of simulation. At the Mol 2 experiments the temperatures (600 and 700°C) and the pressures (up to several hundreds of atmospheres) are exactly controlled and recorded. Thus any fission gas pressure at high burnup and the wall temperature of a fuel pin can be simulated but not the—nevertheless rather important—temperature gradient across the wall. The whole equipment for this series of irradiations was specially designed and is presently installed at the BR 2 site. The pressures are predetermined following two points of view. A part of the tubes will be pressurized so that they should burst in-pile after some time. Hereby clear creep rupture results are wanted. The other part of the specimens will be subjected to lower pressure in order to study the in-pile secondary creep of thin walled tubes. We can reach within reasonable time an integrated fast flux of 2×10^{21} nvt, this order of magnitude being still too low for simulation of real fast reactor fuel pins.

(C) In the materials surveillance subassembly of the EFFBR there are, at present, thirty-two material specimens under irradiation at 350°C in direct contact to the Fermi sodium. This first set of tensile specimens (16Cr13Ni, Inconel 625, V–5Ti–20Nb, V–20Ti–20Nb) will be removed in summer 1966, having then an integrated fast flux of about 5×10^{21} nvt. It will immediately be replaced by a second set of specimens which will, it is hoped, achieve an integrated flux far above 10^{22} nvt. The post-irradiation examination will be the same as in the Mol 1 series. As we have an integrated flux representative here of fast reactor conditions, this experiment is the missing link in the chain and allows control and proof of the extrapolation of former results. Furthermore, the possible influence of fast (n,α) reactions which become evident only with such high integrated fluxes (because of their low neutron cross-sections) can be evaluated.

5. Conclusions

In view of the investigations and available results our preliminary conclusions with reference to the requirements are as follows:

1. With oxide fuel the required linear rod power can easily be achieved below central melting. Special provisions such as Na-bonding, metallic additives, substoichiometry and highest fabricable densities for improving the thermal conductance and conductivity are not necessary.

2. The fuel pins with the proper fuel geometry and density can routinewise be fabricated. The specific requirements have no important influence on pin fabrication economy.

3. The necessary U–Pu homogeneity in the fuel can be achieved by mechanical mixing of the oxide components. The upper limit for PuO_2 particle size of about 150 μ raises no fabrication problem.

4. In a pin type for sodium cooling no special problems arise with respect to the required cladding temperature of 650°C provided that a fission gas space (calculated for 100% fission gas release) is available which limits the maximum pressure to about 100 atm. Also in view of the thermal stresses austenitic steels of the type 16Cr13Ni are sufficient.

5. For steam cooling the necessary maximum temperature of up to 750°C and the high external pressure cause additional problems. Ni-base alloys such as Inconel 625 seem just to meet the requirements if also enough fission gas space is available. A special danger to the stability of the can may arise from creep-buckling. As we consider the ceramic fuel to give no internal support to the wall we might have to provide an extra internal gas pressure.

6. The irradiation damage to can materials of the selected types at doses up to 2×10^{21} n/cm^2 (achieved up to now) are not harmful to mechanical properties. An extrapolation to higher doses seems possible so that there might be no limitations for very high pin burnups.

7. Considering the fuel the required high burnup seems to be achievable provided that no major mechanical fuel clad interaction—due to swelling—takes place. In order to adjust the internal geometry to the expected swelling a high free volume of about 15% in the fuel region is necessary. A too low fuel density, however, might increase the danger of fuel slumping. A high fission gas release is considered to be advantageous as it lowers the swelling rate.

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